Federated Learning for Privacy-Preserving Fleet-Wide Predictive Maintenance Across Distributed Factories

Ananya Gupta, Kenji Tanaka

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ABSTRACT

Cross-enterprise collaborative model training is hindered by strict commercial confidentiality regarding production volumes and failure rates. We formulate Fed-Predict, an asynchronous federated learning framework with differential privacy for fleet-wide predictive maintenance. Across 12 simulated industrial facilities, Fed-Predict achieves 89.3% remaining useful life (RUL) estimation accuracy. However, transmitting large model checkpoints creates severe communication bottlenecks on bandwidth-limited cellular factory uplinks, and non-IID sensor distributions across distinct ambient climates induce significant client drift.

1. Introduction

Original Equipment Manufacturers (OEMs) seek to leverage telemetry from machines sold across worldwide client sites. Centralizing raw sensor streams exposes proprietary production cadences. Federated learning provides a mathematical guarantee of data isolation.

2. Fed-Predict Architecture

Each factory trains local LSTM autoencoders on private telemetry. Local gradients are perturbed with (epsilon, delta)-differential privacy and aggregated via an asynchronous coordinate-median server.

3. Experimental Evaluation

Fleet-wide federated aggregation enables multiple manufacturing plants to collaboratively train predictive maintenance models without sharing proprietary machine telemetry or operational schedules. Fed-Predict improved RUL accuracy by 14.8% over single-plant isolated models.

4. Practical Bottlenecks & Open Questions

Nevertheless, the communication bandwidth required for transmitting 250MB parameter updates over industrial cellular uplinks creates severe synchronization bottlenecks. Most crucially, client non-IID data distribution remains an unsolved roadblock when machine wear rates vary across environmental temperature zones (e.g., Scandinavian vs equatorial production plants).

5. Future Directions

Federated gradient compression and personalized local meta-learning represent essential next steps to handle heterogeneous operating environments.